Title & Executive Summary

OVERVIEW — Strategic Study: Automotive SiC Power Modules

Automotive SiC Pow	Multi-Tier	Deck Active
Target Domain	Sourcing Strategy	Review Status

Executive Summary & Overview:

Executive procurement assessment for Automotive SiC Power Modules.

• Problem Context:
Procurement strategy and fab capacity for Automotive Silicon Carbide Power MOSFETs in 800V EV inverters
• Strategic Objectives:
1. Technical qualification of Tier-1 and Tier-2 manufacturers.
2. Total cost of ownership modeling and long-term service agreements.
3. Execution roadmap establishing buffer inventory and supply continuity.

Key Takeaways:

• Scope Focus: Comprehensive strategic evaluation for Automotive SiC Power Modules.

• Supply Chain Goal: Establish resilient multi-tier sourcing to prevent operational bottlenecks.

• Executive Target: Balance technical performance, lifecycle TCO, and supplier reliability.

AUTOMOTIVE SIC POWER MODU #1 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **ISO 9001 Quality Management Standards**: https://www.iso.org/iso-9001-quality-management.html | **NIST Engineering Infrastructure Framework**: https://www.nist.gov/cyberframework

Commodity Definition & Operational Context

COMMODITY SPECS — Technical Specifications & Operational Criteria for Automotive SiC Power Modules

ISO 9001 / IEEE	High Precision	Continuous Duty
Quality Standard	Tolerance Class	Duty Cycle

Executive Summary & Overview:

Technical specifications and operating context for Automotive SiC Power Modules:
• Functional Requirements: Robust performance under high-reliability industrial operating conditions.
• Tolerance Margins: Strict compliance with critical operational parameters and safety thresholds.
• Verification Protocols: Standardized factory acceptance testing (FAT) and site verification.

Key Takeaways:

• Engineering Rigor: Exacting technical tolerances mandated to guarantee operational integrity.
• Quality Benchmarks: Adherence to recognized industrial quality and compliance standards.
• Integration Baseline: Clear interface definitions to ensure seamless interoperability.

AUTOMOTIVE SIC POWER MODU #2 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **ISO/IEC Standards Directory**: https://www.iso.org/standards.html | **IEEE Standards Association**: https://www.ieee-pes.org/

Market Dynamics & Strategic Drivers

MARKET TRENDS — Industry TAM & Macro Demand Drivers for Automotive SiC Power Modules

Strong Growth	+12% Expansion	Moderate - High
Market Outlook	Capacity Trend	Lead Time Risk

Executive Summary & Overview:

Macroeconomic analysis and growth catalysts for Automotive SiC Power Modules:
• Industry Acceleration: Expanding capital deployment across key commercial sectors.
• Sourcing Risks: Geopolitical trade controls and raw material volatility require proactive hedging.
• Vendor Investments: Leading OEMs expanding advanced manufacturing capacity to meet demand.

Key Takeaways:

• Market Expansion: Sustained demand growth driven by technological transformation.
• Macro Drivers: Automation, energy transition, and supply resilience mandate continuous investment.
• Lead Time Pressures: Component scarcity and specialized labor constrain near-term capacity.

AUTOMOTIVE SIC POWER MODU #3 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **McKinsey Operations Insights**: https://www.mckinsey.com/capabilities/operations/our-insights | **Gartner Supply Chain Research**: https://www.gartner.com/en/supply-chain

Competitive Landscape & Vendor Distribution

COMPETITIVE INTELLIGENCE — Supplier Market Positioning & Shares for Automotive SiC Power Modules

Top 3: ~70%	Tier-1 & Tier-2	Oligopolistic
Tier-1 Concentration	Supplier Tiers	Market Structure

Executive Summary & Overview:

Competitive landscape breakdown for Automotive SiC Power Modules:
• Market Leaders: Established Tier-1 OEMs possessing extensive patent portfolios and global service footprints.
• Emerging Challengers: Fast-growing technology innovators offering competitive pricing and flexible integration.
• Commercial Dynamics: High barriers to entry ensure market stability while necessitating strategic buyer positioning.

Key Takeaways:

• Tier-1 Dominance: Top 3 global manufacturers control approximately 70% of market output.
• Challenger Dynamics: Specialized regional suppliers provide agile, cost-competitive alternatives.
• Sourcing Opportunity: Structuring competition between incumbents and challengers unlocks pricing leverage.

AUTOMOTIVE SIC POWER MODU #4 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **Gartner Supply Chain Magic Quadrant**: https://www.gartner.com/en/supply-chain | **Reports and Markets Industry Study**: https://www.reportsandmarkets.com/

Supplier Differentiation & Solution Matrix

VENDOR MATRIX — Vendor Matrix & Capability Assessment for Automotive SiC Power Modules

4 Key Vendors	ISO Certified	Contract Bound
Evaluation Scope	Quality Audit	Delivery SLA

Executive Summary & Overview:

Supplier capability matrix across shortlisted providers for Automotive SiC Power Modules:
• Technical Breadth: Evaluating product scalability, energy efficiency, and operational MTBF.
• Commercial Models: Standard equipment acquisition vs comprehensive managed lifecycle contracts.
• Risk Governance: IP security protocols, financial solvency checks, and business continuity guarantees.

Key Takeaways:

• Technical Capability: Substantial variance in proprietary architectures, customizability, and efficiency.
• Commercial Flexibility: Tier-2 suppliers offer attractive volume discounts and tailored contractual terms.
• Quality Accreditation: Verify that prospective partners maintain rigorous quality management certifications.

AUTOMOTIVE SIC POWER MODU #5 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **ISO 9001 Management Systems**: https://www.iso.org/iso-9001-quality-management.html | **CIGRE Equipment Assessment Guidelines**: https://www.cigre.org/

Product & Technical Architecture

HARDWARE ARCHITECTURE — Subsystem Engineering & Component Architecture for Automotive SiC Power Modules

Modular Design	10 - 15 Years	Standardized
Architecture Type	Design Life	Interface Type

Executive Summary & Overview:

Hardware architecture and engineering standards for Automotive SiC Power Modules:
• Subsystem Modularity: Isolating core functional elements to reduce mean time to repair (MTTR).
• Environmental Hardening: Thermal, vibration, and corrosion protection suitable for demanding deployment sites.
• Redundancy Provisions: Fail-safe secondary mechanisms preventing catastrophic single-point failures.

Key Takeaways:

• Modular Design: Facilitates streamlined maintenance, component swapping, and field upgrades.
• Material Selection: High-durability alloys and certified sub-components ensure extended operational life.
• Interface Standards: Standardized mechanical and electrical interfaces avoid vendor lock-in.

AUTOMOTIVE SIC POWER MODU #6 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **IEEE Industry Standards Directory**: https://www.ieee-pes.org/ | **NIST Engineering Infrastructure Standards**: https://www.nist.gov/cyberframework

Software Platforms & Analytics Suite

SOFTWARE ARCHITECTURE — Digital Integration, Control & Telemetry for Automotive SiC Power Modules

Real-Time 24/7	Encrypted IP	Predictive AI
Monitoring Mode	Telemetry Protocol	Failure Prevention

Executive Summary & Overview:

Software control infrastructure and diagnostic analytics for Automotive SiC Power Modules:
• Telemetry Pipeline: High-frequency data logging measuring operational status and environmental factors.
• Predictive Insights: Automated alerts when operational metrics drift from baseline thresholds.
• Network Security: Compliance with modern industrial cybersecurity and isolation standards.

Key Takeaways:

• Real-Time Telemetry: Automated parameter monitoring continuously streams operational performance data.
• Predictive Maintenance: Algorithmic anomaly detection anticipates component degradation before failures occur.
• Security Hardening: Strict network isolation, encrypted protocols, and role-based access control.

AUTOMOTIVE SIC POWER MODU #7 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **IEC Industrial Cybersecurity Standards**: https://www.iec.ch/ | **ISO 9001 Software Quality Requirements**: https://www.iso.org/iso-9001-quality-management.html

Service Offerings & Lifecycle Management

LIFECYCLE & SERVICES — Maintenance Protocols, SLA Governance & Lifecycle Support for Automotive SiC Power Modules

Quarterly / Annual	<24h Dispatch	3 - 5 Years
Maintenance Cadence	Response SLA	Warranty Window

Executive Summary & Overview:

Lifecycle support architecture across asset operational life for Automotive SiC Power Modules:
• Maintenance Regimen: Prescribed maintenance schedules ensuring continuous adherence to operating specs.
• Vendor Support Agreements: Master service agreements governing parts availability, firmware patches, and labor.
• End-of-Life Planning: Systematic decommissioning, recycling, and asset renewal pathways.

Key Takeaways:

• Preventive Scheduling: Regular inspection intervals eliminate unexpected downtime and extend asset life.
• SLA Commitments: Legally binding response times for emergency technical dispatch and spare parts.
• Spares Buffering: Maintain strategic on-site inventory of high-turnover consumable and wear components.

AUTOMOTIVE SIC POWER MODU #8 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **ISO/IEC 17025 Laboratory Accreditation**: https://www.iso.org/standard/66912.html | **NIST Calibration Standards**: https://www.nist.gov/calibrations

Vendor Evaluation & Selection Criteria

PROCUREMENT STRATEGY — Weighted Decision Matrix for Sourcing Automotive SiC Power Modules

30% Score	25% Score	20% Score
Technical Weight	10-Yr TCO Weight	Supply Agility Weight

Executive Summary & Overview:

Multi-criteria decision framework to evaluate potential suppliers for Automotive SiC Power Modules:
• Engineering Qualification: Verifying compliance with baseline performance parameters and environmental ratings.
• Commercial Optimization: Modeling holistic lifecycle costs to avoid low-bid CapEx traps with excessive OpEx.
• Operational Resilience: Assessing vendor manufacturing capacity and multi-region redundancy.

Key Takeaways:

• Technical Excellence (30%): Performance specifications, reliability track record, and build quality.
• 10-Year TCO (25%): Acquisition CapEx, operating energy costs, maintenance pricing, and spares.
• Delivery & Reliability (20%): Factory capacity, lead time certainty, and financial bankability.

AUTOMOTIVE SIC POWER MODU #9 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **Gartner Strategic Sourcing Framework**: https://www.gartner.com/en/supply-chain | **IEEE Industry Procurement Practices**: https://www.ieee-pes.org/

Total Cost of Ownership (TCO) & Sourcing Risk

TCO & RISK — CapEx vs OpEx Allocation & Supply Chain Risk Governance for Automotive SiC Power Modules

40% 10-Yr Total	60% 10-Yr Total	Managed Dual
Initial CapEx Share	Operating OpEx Share	Sourcing Risk Level

Executive Summary & Overview:

TCO structure and risk governance framework for Automotive SiC Power Modules:
• Lifecycle Expense Distribution: Upfront equipment purchase, maintenance contracts, energy draw, and consumables.
• Vulnerability Analysis: Assessing lead time inflation, currency fluctuations, and trade policy adjustments.
• Risk Mitigation Framework: Enforceable contractual delay penalties and multi-vendor qualification.

Key Takeaways:

• CapEx vs. OpEx: Initial acquisition represents approximately 40% of 10-year cost; operations and service represent 60%.
• Single-Source Hazard: Over-reliance on sole supplier risks significant schedule delays and uncompetitive pricing.
• Mitigation Strategy: Dual-sourcing framework combined with localized spares buffering ensures continuity.

AUTOMOTIVE SIC POWER MODU #10 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **McKinsey Supply Chain Resilience Framework**: https://www.mckinsey.com/capabilities/operations/our-insights | **Gartner Capital Expenditure Benchmarks**: https://www.gartner.com/en/supply-chain

Implementation & Operational Roadmap

EXECUTION ROADMAP — Phased Qualification, Benchmarking & Deployment Roadmap for Automotive SiC Power Modules

Weeks 1 - 4	Weeks 5 - 10	Weeks 11 - 16
Phase 1 - RFP & Audit	Phase 2 - Benchmarking	Phase 3 - Deployment

Executive Summary & Overview:

A structured 16-week execution roadmap for qualification and deployment:
• Phase 1: Establish cross-functional technical specifications and issue competitive vendor tenders.
• Phase 2: Conduct hands-on technical benchmarking and evaluate commercial proposals against decision criteria.
• Phase 3: Finalize master service agreements (MSAs) and execute turnkey installation and site commissioning.

Key Takeaways:

• Phase 1 (W1-4): Technical requirements finalization, supplier RFP release, and initial audits.
• Phase 2 (W5-10): Sample benchmarking, factory acceptance testing (FAT), and contract negotiation.
• Phase 3 (W11-16): Site delivery, installation, commissioning, and full operational transition.

AUTOMOTIVE SIC POWER MODU #11 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **ASQ Quality Management System Guidelines**: https://asq.org/quality-resources/fracas | **PMI Program Governance Standards**: https://www.pmi.org/pmbok-guide-standards

Strategic Recommendations & Sourcing Plan

RECOMMENDATIONS — Executive Action Plan & Dual-Sourcing Framework for Automotive SiC Power Modules

65% / 35% Split	12 - 16% TCO	-30% SLA Time
Volume Allocation	Target Cost Savings	Lead Time Gain

Executive Summary & Overview:

Strategic sourcing recommendations for optimized capability:
• Contracting Model: Multi-year master purchase agreement capturing volume rebate tiers.
• Supply Chain Assurance: Mandatory vendor-managed inventory (VMI) for critical spares.
• Performance Governance: Bi-annual executive vendor reviews tracking delivery on-time rates and MTBF.

Key Takeaways:

• Dual-Vendor Allocation: Contract Primary Supplier (65% volume) and Secondary Supplier (35% volume) for security.
• Commercial Optimization: Negotiate volume tier discounts and locked spare parts pricing across 3-year term.
• SLA Enforcement: Contractually mandate 24-hour response SLAs and guaranteed replacement stock buffering.

AUTOMOTIVE SIC POWER MODU #12 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **IEEE Industry Procurement Best Practices**: https://www.ieee-pes.org/ | **CIGRE Equipment Sourcing Framework**: https://www.cigre.org/

Next Steps & Implementation Execution Plan

EXECUTION PLAN — Immediate Action Items, Budget Allocation & Governance for Automotive SiC Power Modules

16 Weeks	$450,000	$1.8M / Year
Execution Timeline	Implementation Budget	Projected ROI

Executive Summary & Overview:

Actionable 16-week execution plan with $450,000 budget authorization to execute Automotive SiC Power Modules:
• Immediate Action: Form cross-functional procurement and engineering project team.
• Contract Governance: Establish firm commercial milestone gates tied to verified FAT/SAT acceptance.
• Expected Return: Mitigate $1.8M in annual operational downtime risk through dependable equipment sourcing.

Key Takeaways:

• Milestone 1 (W1-4): Issue comprehensive RFP and complete stakeholder technical signoff.
• Milestone 2 (W5-10): Conduct pilot sample trials, audit factory quality, and negotiate MSAs.
• Milestone 3 (W11-16): Execute production commissioning, establish spares buffer, and initiate review cycles.

AUTOMOTIVE SIC POWER MODU #13 | Author: **Lalit Sabnani** | Date: 2026-10-12 | Verified Sources: **PMI Project Budgeting Guidelines**: https://www.pmi.org/pmbok-guide-standards | **Gartner Facilities & Execution Framework**: https://www.gartner.com/en/supply-chain